

Plotting Air Quality Index Data on a Map Using Python

Rising air pollution levels in cities around the world are a matter of concern. This script uses two Python libraries — Pandas and Folium — to plot the air quality index (AQI) for India.

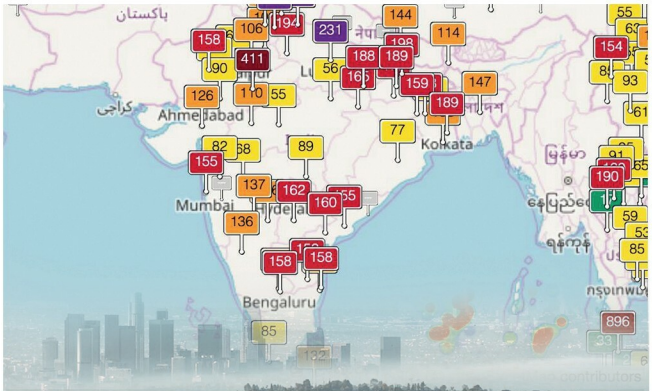


Image source: <https://medium.com>

This project uses two Python libraries. The first is Pandas, which is a well-known library for data science. But it also has capabilities for downloading data and even plotting data. The second library is Folium, which is a 'Python port' of an excellent JavaScript library called Leaflet.

For better understanding, this script is written in the steps given below.

1. Download the air quality index (AQI) data. This step consists of the following sub-steps:
 - Get a token from the aqicn.org website.
 - Understand how to *construct* the API for downloading data.
 - Create the *bounding box* for which the AQI data is to be downloaded.

- Use the `read_json()` function of Pandas to make an API call.
2. Create a Pandas DataFrame from the downloaded data.
 3. Clean the data.
 4. Create a heat map.
 5. Plot the measuring stations on the Folium map.

Step 1: Downloading AQI data

The data source for this script is the World Air Quality Index project at <https://aqicn.org/>. The website provides many different APIs to download data. However, to do this, you need an access token, for which you need to register on the website with your email and name. Once you do the token will be delivered to your mail ID almost immediately.